

Dr. Sunilgar Laxmangar Gusai

Assistant Professor,

Mathematics for Computer Science, AI & Data Science

Specializing in spectral graph theory, mathematical foundations of machine learning, and interdisciplinary applications of mathematics in data science, network analysis, and computational systems.



Ph.D. in Mathematics



GSET (Mathematical Sciences)



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Portfolio



LinkedIn

Academic Profile

- Emerging researcher in **spectral graph theory** with a focus on **graph energy measures**, vertex-eccentricity based invariants, and mathematical modeling of complex networks. Developed the concept of **Vertex Eccentricity Labeled Energy (VELE)** and investigated its structural behavior across standard graph families and graph operations.
- Research work extends towards **network analysis, QSPR modeling, and data-driven mathematical frameworks**, with applications in mathematical chemistry and structural resilience of networks.
- Actively engaged in **teaching undergraduate and postgraduate students across computing and mathematics programs (BCA, B.Sc. IT, B.Sc. Data Science, M.Sc. Data Science, M.Sc. Mathematics, and MCA)**, with emphasis on delivering core mathematical foundations—linear algebra, probability, statistics, and optimization—aligned with their applications in **data science, machine learning, and computational problem solving**.
- Research Interests:** Graph Theory and Network Analysis; Spectral Graph Theory; Graph Energy and Eccentricity-Based Graph Invariants; Quantitative Structure–Property Relationships (QSPR) and Mathematical Chemistry; Mathematical Foundations of Data Science; Linear Algebra and Optimization; Mathematical Modeling and Computational Methods; Graph Neural Networks (GNNs) and Network-Based Machine Learning; Emerging Directions in Quantum Computing; Interdisciplinary Applications of Mathematics in Computing and Complex Systems.

Educational Qualification

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| Jan 2026 | Ph.D. in Mathematics , Saurashtra University, Rajkot.
Thesis title: <i>Discussion on Some Topics of Current Interest in Graph Theory</i> . |
| Mar 2020 | Gujarat State Eligibility Test (GSET) , Mathematical Sciences.
Qualified State Eligibility Test for Assistant Professor during M.Sc.. |
| 2018 – 2020 | M.Sc. in Mathematics , Krantiguru Shyamji Krishna Verma Kachchh University, Kachchh, Gujarat. Percentage: 78.33% . |
| 2015 – 2018 | B.Sc. (Hons.) in Mathematics , Krantiguru Shyamji Krishna Verma Kachchh University, Kachchh, Gujarat. Percentage: 78.00% . |
| 2015 | Higher Secondary Certificate (H.S.C.) , Gujarat Secondary and Higher Secondary Education Board. Percentage: 73.85% . |
| 2013 | Secondary School Certificate (S.S.C.) , Gujarat Secondary and Higher Secondary Education Board. Percentage: 85.33% . |

Experience

- Dec 2025 – Present ■ **Program Coordinator – Data Science Programs, Marwadi University.**
Serving as Program Coordinator for the **B.Sc. Data Science** and **M.Sc. Data Science** programmes. Responsibilities include academic planning, curriculum coordination, course scheduling, faculty coordination, student mentoring, and academic administration. Contributing to the development and delivery of interdisciplinary curriculum integrating **mathematics, data science, and computational methodologies**, along with support for accreditation and quality assurance processes.
- Aug 2022 – Present ■ **Assistant Professor of Mathematics, Faculty of Computer Applications, Marwadi University, Rajkot.**
Teaching undergraduate and postgraduate students across programmes including **BCA, B.Sc. IT, B.Sc. Data Science, M.Sc. Data Science, M.Sc. Mathematics, and MCA**. Courses taught include **Linear Algebra, Calculus, Discrete Mathematics, Operations Research, Probability Theory, Statistics, and Applied Mathematics**, with emphasis on their applications in **data science, machine learning, and computational problem solving**. Actively engaged in research, academic mentoring, curriculum development, and interdisciplinary collaboration between mathematics and computing domains.
- Aug 2023 – Present ■ **Coordinator, Time Table Committee, Marwadi University, Rajkot.**
Involved in academic scheduling, coordination of departmental timetables, documentation, and institutional academic administration, ensuring smooth academic operations.
- Jan 2021 – Apr 2022 ■ **Assistant Professor (Contractual), Department of Mathematics, Saurashtra University, Rajkot.**
Delivered undergraduate and postgraduate mathematics courses, contributed to student mentoring, and supported academic and departmental activities.

Academic Leadership and Administrative Contributions

- 2023 – Present ■ **Academic Administration and Quality Support.**
Actively involved in academic planning, curriculum structuring, examination work, student mentoring, documentation, committee coordination, and institutional processes related to quality assurance and accreditation.
- 2024 ■ **Departmental Conference Support.**
Contributed to coordination and academic support for an international conference organized by the department at Marwadi University.
- 2026 ■ **Industrial Visit Coordination.**
Organized and coordinated an industrial visit to **BISAG (Bhaskaracharya National Institute for Space Applications and Geo-informatics), Gandhinagar**, facilitating industry exposure for students in data science and technology domains.
- 2025 ■ **Industrial and Educational Visit Coordination.**
Organized and coordinated an academic industrial visit for students to leading technology and networking organizations including **Network Bulls Technologies, Gurugram** and other IT industry exposure centres in North India, facilitating practical understanding of networking, cybersecurity, and emerging computing technologies. The visit also included educational and cultural exposure at nationally significant locations such as **Agra and Ayodhya**.

Publications

Published Research Articles

- 2026  **Eccentricity-Based Bounds for the Spectral Radius of Graph Matrices.**
International Journal of Science and Research (IJSR), Vol. 15, pp. 681–686, 2026. ISSN: 2319-7064 (Online).
- 2025  **On Vertex Eccentricity Labeled Energy of a Graph.**
International Journal of Basic and Applied Sciences, Vol. 14(4), pp. 339–350, 2025. **Scopus Indexed.**
DOI: <https://doi.org/10.14419/qtfv0860>

Manuscripts Communicated / Under Review

-  **Vertex Eccentricity Labeled Energy of Graphs Obtained from Graph Operations.**
Manuscript under review.
-  **Signless Laplacian Energy of a Graph with Self Loops.**
Manuscript under review.
-  **Tiered QSPR Modeling of Benzyl Sulfide Derivatives Integrating Graph-Theoretic and Polarity Descriptors: Linear, Quadratic, and Descriptor-Reduced Multivariate Approaches.**
Manuscript under review.
-  **VELE-Based Structural Resilience Analysis in Networks.**
Manuscript under development for submission to a high-impact journal.

Work in Progress

- 2026  **QSPR Modeling of Protein Structures Using Graph-Theoretic Descriptors.**
Work in progress.
-  **Advanced Spectral Descriptors for Complex Network Analysis.**
Work in progress.

Research Project

- 2024 – Present  **Seed Grant (Minor Research Project)**, Marwadi University, Rajkot.
Title: *The Study on Variants of Graph Energy.*
Role: Principal Investigator (PI).
Funding: Rs. 80,000 (Internal Research Grant).
Duration: April 1, 2024 – Present.
Research focused on **spectral graph theory and graph energy measures (including VELE)**, with applications in **network analysis, structural properties of graphs, and data-driven mathematical modeling.**

Paper Presentation

- Nov–Dec 2023  **A Study on Vertex Eccentricity Labeled Energy of a Graph.**
Presented at the *International Conference on Graph Theory and its Applications (ICGTA 2023)*, CHRIST (Deemed to be University), Bangalore, India, 30 November – 02 December 2023.
- Feb 2024  **On Vertex Eccentricity Labeled Energy of a Graph.**
Presented at the *2nd International Conference on Emerging Trends & Contemporary Practices (ICETCP 2024)*, Atmiya University, Rajkot, India, 09–10 February 2024.

Participations: FDP / STTP / Workshop / Seminar / Certifications

Faculty Development Programmes (FDP)

- 1 Participated in the *Two-Week Online Faculty Development Programme on “Next-Generation Data Science and Artificial Intelligence: Industry-Driven Technologies and Solutions (DSAI-2026)”*, organized by Annant Gyan Knowledge & Skills Pvt. Ltd., from 16 March 2026 to 28 March 2026.
- 2 Participated in the *Five-Day Online Faculty Development Programme on “Graph-Based Machine Learning and Neural Networks: Theory, Models, and Applications”*, organized by Annant Gyan Knowledge & Skills Pvt. Ltd., from 27/01/2026 to 31/01/2026.
- 3 Participated in the *One Week National Level Online Hands-On FDP on “Data Analytics with Mathematics & Python”*, organized by Indian Mathematics Professors Association and Infinity Research & Development Institute, Coimbatore, from 24/11/2025 to 28/11/2025.
- 4 Participated in the *One Week National Level Online FDP on “Mastering Operations Research with Advanced Tools: Python, SageMath, MATLAB & LaTeX Draw”*, organized by Indian Mathematics Professors Association and Infinity Research & Development Institute, Coimbatore, from 23/06/2025 to 28/06/2025.
- 5 Participated in the *Five-Day Virtual Faculty Development Programme on “Interdisciplinary Research Collaborations and Innovative Teaching Methods for Mathematics”*, organized by Department of Mathematics in association with IQAC and IIC, St. Joseph’s College (Arts & Science), from 17/03/2025 to 21/03/2025.
- 6 Participated in the *Six Days International Faculty Development Program on “Empowering Advancement in AI for Education, Healthcare and Finance”*, organized by Department of Computer Applications (MCA), Hindusthan College of Engineering and Technology, Coimbatore, from 10/03/2025 to 15/03/2025.
- 7 Completed the *NPTEL Faculty Development Programme (FDP) equivalent course on “Advanced Graph Theory”*, IIT Kanpur (SWAYAM), Jan–Mar 2025, with consolidated score **75% (Elite + Silver)**.
- 8 Participated in the *One Week Faculty Development Programme on “Statistical Applications and Data Analytics for Modern Research”*, organized by Department of Mathematics, SRM Institute of Science and Technology, Vadapalani Campus, Chennai, from 17/12/2024 to 22/12/2024.
- 9 Participated in the *Five Day Online Faculty Development Programme on “Research in Mathematics: Present Trends and Future Prospects”*, organized by PG and Research Department of Mathematics, Sacred Heart College (Autonomous), Tirupattur, in collaboration with Patrician College of Arts and Science, Chennai, from 18/11/2024 to 22/11/2024.
- 10 Participated in the *One Week International Faculty Development Program on “Essential Mathematics for Machine Learning, Deep Learning and Cyber Security”*, organized by Department of Mathematics, SRM Institute of Science and Technology, Vadapalani Campus, Chennai, from 18/10/2024 to 25/10/2024.
- 11 Participated in the *Five Day Faculty Development Programme on “Advances in Graph Theory and Linear Algebra (AGTLA)”*, organized by Department of Mathematics, School of Advanced Sciences, VIT–AP University, from 01/10/2024 to 05/10/2024.
- 12 Participated in the *Five Day National Virtual Faculty Development Programme on “Data Driven Decision Making: Applied Linear Algebra, Integrating Statistical Methods and Data Science in Research”*, organized by Department of Mathematics in association with IQAC & IIC, St. Joseph’s College (Arts & Science), from 23/09/2024 to 27/09/2024.
- 13 Completed the *NPTEL Faculty Development Programme (FDP) equivalent course on “Introduction to Abstract and Linear Algebra”*, IIT Kharagpur (SWAYAM), Jul–Sep 2024, with consolidated score **80% (Elite + Silver)**.

- 14 Participated in the *One-Week National Level Faculty Development Programme (Hybrid Mode) on "Assimilation of Indian Knowledge Systems with NEP-2020: Prospect and Retrospect"*, organized by Department of Basic Sciences & Humanities and IQAC, BVRIT Hyderabad College of Engineering for Women, from 30/07/2024 to 04/08/2024.
- 15 Participated in the *Faculty Development Program on "Real-Life Applications of Mathematics and Actuarial Science"*, organized by Department of Mathematics and Actuarial Science, B. S. Abdur Rahman Crescent Institute of Science and Technology, from 25/07/2024 to 30/07/2024.
- 16 Participated in the *One Week International FDP on "Numerical Computations and Dynamical Systems (NCDS-2024)"*, organized by Department of Mathematics, Anurag University, Hyderabad, from 22/07/2024 to 26/07/2024.
- 17 Participated in the *One Week Faculty Development Programme on "Mathematical Foundation for AI, ML & Data Science"*, organized by Department of Mathematics, SRM Institute of Science and Technology, Vadapalani Campus, Chennai, from 11/06/2024 to 19/06/2024.
- 18 Participated in the *Five Day International Online FDP on "Exploring Recent Trends in Applied Mathematics and Machine Learning"*, organized by Department of Mathematics, Koneru Lakshmaiah Educational Foundation, Hyderabad, from 20/05/2024 to 24/05/2024.
- 19 Participated in the *Six Days National Level Faculty Development Programme on "AI in Learning, Teaching, Assessment and Research Practices"*, organized by Department of Mathematics, Bon Secours College for Women, Thanjavur, from 06/05/2024 to 11/05/2024.
- 20 Participated in the *Five Days FDP on Concept of Pure and Applied Mathematics*, jointly organized by Parul University and Gujarat Ganit Mandal, from 11/03/2024 to 15/03/2024.
- 21 Participated in the *Five-Day Online FDP on "Advances in Graph Theory"*, organized by Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, from 11/09/2023 to 15/09/2023.

Short Term Training Programme (STTP)

- 1 Participated in the *Online Short-Term Training Program on "Unlocking Potential: Advanced Mathematics for Engineers and Scientists"*, organized by Parul University in collaboration with Department of Mathematics, NIT Jamshedpur, from 21/10/2024 to 26/10/2024.
- 2 Participated in the *Seven Day Short-Term Training Programme (Virtual) on "MATLAB"*, organized by Department of Mathematics, CSSR & SRRM Degree & PG College, Kamalapuram, from 08/04/2024 to 14/04/2024.

Workshops / Training

- 1 Participated in the training workshop *"Transforming Higher Education with Google Gemini"*, in collaboration with STTAR, on 19/12/2025.
- 2 Participated in the *One-Day Workshop on "LaTeX for Scientific Documentation"*, organized by MathTech Thinking Foundation, on 29/12/2024.
- 3 Participated in the *One-Day Workshop on "Computing Calculus with Python"*, organized by MathTech Thinking Foundation, on 23/12/2024.
- 4 Participated in the *National Workshop on Computational Linear Algebra and Graph Theory (NWCLAGT 2024)*, organized by CARAMS, MAHE, Manipal, from 18/12/2024 to 20/12/2024, and earned one credit.
- 5 Completed the *2-Day Workshop on "Business Intelligence using Power BI"*, conducted by Skill Nation, on 04/05/2024 and 05/05/2024.

- 6 Participated in the *One Week National Workshop (Virtual) on "SCILAB"*, organized by Department of Mathematics, CSSR & SRRM Degree & PG College, Kamalapuram, from 29/04/2024 to 04/05/2024.

Conference / Seminar Participation

- 1 Participated in the *Online Foreign Guest Lecture on "Wiener Index of Graphs"* delivered by Dr. Dinesh Pandey (Wilfrid Laurier University, Canada), organized by VIT-AP University, on 29/07/2025.
- 2 Participated in the *National Seminar on "Recent Trends in Mathematics"*, organized by Department of Mathematics, Maharaja Krishnakumarsinhji Bhavnagar University, Bhavnagar, on 11/03/2024.
- 3 Participated in the *International Conference on New Age Mathematics for Data Science and Artificial Intelligence (ICNAMDA-2024)*, organized by Department of Mathematics, Rathinam College of Arts and Science, Coimbatore, on 01/03/2024.
- 4 Participated in the *International Conference on Applied Mathematics and Nanoscience (ICAMN-2024)*, organized by Vidya Academy of Science and Technology, Thrissur, Kerala, from 23/01/2024 to 25/01/2024.
- 5 Participated in the *International Conference on Recent Advancements in Graph Theory (ICRAGT-2022)*, organized by Department of Mathematics, Gujarat University, from 10/09/2022 to 11/09/2022.
- 6 Participated in the *18th Annual Conference of ADMA and Graph Theory Day*, organized by Department of Mathematics, Vellalar College for Women (Autonomous), Erode, Tamil Nadu, from 09/06/2022 to 11/06/2022.
- 7 Participated in the *National Conference on Emerging Trends in Discrete Mathematics (NCETDM-2022)*, organized by Hemchandracharya North Gujarat University, Patan, in association with Atmiya University, Rajkot, from 17/02/2022 to 19/02/2022.
- 8 Participated in the annual conference of *Gujarat Ganit Mandal*, organized at Ganpat University, Mehsana Urban Institute of Sciences, in December 2016.

Online Courses / Technical Certifications

- 1 Completed the *NPTEL Online Certification Course on "Advanced Graph Theory"*, IIT Kanpur, Jan-Mar 2025, with consolidated score **75% (Elite + Silver)**.
- 2 Completed the *NPTEL Online Certification Course on "Introduction to Abstract and Linear Algebra"*, IIT Kharagpur, Jul-Sep 2024, with consolidated score **80% (Elite + Silver)**.
- 3 Completed the UdeMy certification course "*Google Sheets Bootcamp: Boost Productivity and Efficiency*", September 2024.
- 4 Completed the UdeMy certification course "*Python Development & Data Science: Variables and Data Types*", September 2024.
- 5 Completed the UdeMy certification course "*Introduction to R Programming*", August 2024.
- 6 Completed the UdeMy certification course "*2024 Complete Python Bootcamp from Zero to Hero in Python*", August 2024.
- 7 Completed the UdeMy certification course "*Python Course From Basic to Advanced*", August 2024.
- 8 Completed the UdeMy certification course "*Computer Vision with Python*", August 2024.
- 9 Completed the UdeMy certification course "*NumPy, SciPy, Matplotlib & Pandas A-Z: Machine Learning*", August 2024.

Technical Skills and Competencies

Mathematics	■ Graph Theory; Spectral Graph Theory; Graph Energy and Matrix Analysis; Linear Algebra; Probability and Statistics; Operations Research; Mathematical Modeling and Optimization.
Computing	■ Working knowledge of Python for mathematical and statistical computation; familiarity with data analysis, numerical methods, and algorithmic problem solving.
Tools	■ $\LaTeX$; Microsoft Office Suite (Excel, Word, PowerPoint); exposure to computational and mathematical software for academic teaching and research.
Data Science	■ Mathematical foundations of data science; statistical analysis; exposure to machine learning concepts; application of mathematical methods in data-driven and computational contexts.
Academic	■ Curriculum development; academic planning and coordination; student mentoring; interdisciplinary teaching; academic documentation and quality assurance processes.

Professional Memberships

2026 – Present	■ Association for Computing Machinery (ACM). Professional Member (Membership No.: 0298229). Area: Graph Theory, Network Science, Artificial Intelligence.
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Achievements

2025	■ Kandala Merit Scholarship , awarded under Deendayal Port Trust Merit Scholarship Scheme for securing First Rank in M.Sc. (Mathematics).
2020	■ Qualified GSET (Gujarat State Eligibility Test) , Mathematical Sciences, during M.Sc. (2019), demonstrating strong foundational competence in core mathematical subjects.
2024	■ Appreciation Certificate , Marwadi University, for contribution to Anti-Ragging Awareness Initiative.

References

- **Dr. Vinodray J. Kaneria**
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Department of Mathematics,
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- **Dr. Mitesh J. Patel**
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Tolani College of Arts and Science, Adipur
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Statement of Research

My research interests lie primarily in the area of **Spectral Graph Theory**, with particular emphasis on **graph energy measures**, eccentricity-based graph invariants, and their interdisciplinary applications in computing, network analysis, and data-driven systems. My research aims to connect rigorous mathematical theory with emerging computational and technological domains through structurally meaningful and application-oriented mathematical frameworks.

A significant part of my research work focuses on the development and analysis of **Vertex Eccentricity Labeled Energy (VELE)**, a graph energy invariant based on eccentricity-labelled graph matrices. My work investigates its structural behaviour across standard graph families and graph operations, along with its spectral properties, mathematical characteristics, and potential applications in complex network analysis. This research contributes to the broader development of graph energy theory and spectral methods in graph analysis.

Beyond theoretical graph theory, my research extends towards **network analysis, data science, and computational modeling**. I am particularly interested in studying how graph-theoretic descriptors and spectral techniques can support the analysis of communication networks, structural resilience, and computational systems. I have also been actively involved in interdisciplinary research related to **Quantitative Structure-Property Relationship (QSPR) modeling**, where graph-based mathematical descriptors are utilized for applications in mathematical chemistry and cheminformatics.

My emerging research directions include applications of graph-based methodologies in **graph neural networks (GNNs)**, machine learning, and computational intelligence. I am interested in exploring how mathematical structures from graph theory can contribute to network-oriented learning systems and modern AI frameworks. Additionally, I have growing interest in advanced computational paradigms such as **quantum computing** and mathematically driven approaches for complex computational systems.

I strongly believe in interdisciplinary and collaborative research that connects mathematics with computer science, artificial intelligence, data science, and applied computational domains. My long-term research objective is to develop mathematically grounded methodologies that contribute both to theoretical advancement in graph theory and to meaningful real-world computational applications. In the long term, I aim to contribute towards interdisciplinary research connecting mathematics, artificial intelligence, data science, and network-oriented computational systems through mathematically rigorous and application-driven approaches.

Statement of Teaching

My teaching philosophy is centered on building strong conceptual understanding while connecting mathematics with practical, computational, and interdisciplinary applications. I believe that mathematics forms the foundational language of modern computing disciplines; therefore, effective teaching should balance theoretical rigor with meaningful applications in areas such as computer science, data science, artificial intelligence, machine learning, and computational modeling.

I have experience teaching undergraduate and postgraduate students across programmes including **BCA, MCA, B.Sc. IT, B.Sc. Data Science, M.Sc. Data Science, and M.Sc. Mathematics**. The courses I teach include **Linear Algebra, Calculus, Discrete Mathematics, Operations Research, Probability Theory, Statistics, and Applied Mathematics**. In my classroom approach, I emphasize conceptual clarity, analytical reasoning, and problem-solving skills while demonstrating how mathematical ideas are applied in computational and data-driven environments.

I strongly believe that students learn mathematics more effectively when they understand its relevance to real-world and technological problems. Therefore, I frequently integrate examples related to **data analysis, optimization, network analysis, algorithmic thinking, and machine learning foundations** into teaching discussions. My objective is not only to teach mathematical methods, but also to help students develop confidence in applying mathematical thinking to interdisciplinary and emerging domains.

In addition to classroom teaching, I actively contribute to curriculum development, academic mentoring, programme coordination, and institutional academic activities. Through my role as **Program Coordinator for Data Science Programmes**, I have been involved in academic planning, curriculum structuring, student mentoring, interdisciplinary academic coordination, and support for quality assurance processes. These experiences have strengthened my understanding of outcome-based education and the evolving academic needs of computing-oriented programmes.

I also consider continuous learning and academic engagement to be an essential part of effective teaching. To enhance both subject knowledge and pedagogy, I regularly participate in faculty development programmes, workshops, and interdisciplinary academic activities related to mathematics, data science, artificial intelligence, and computational technologies.

My overall goal as an educator is to help students build strong mathematical foundations, intellectual curiosity, and analytical maturity while encouraging them to apply mathematical thinking in modern technological and research environments. I aspire to create a learning atmosphere where students can confidently connect mathematical concepts with computational challenges and interdisciplinary innovation.